Why a cover, an isogeny or a Jacobian chain does not move an elliptic-curve discrete log: the obstruction, stated once

2026-11-12

Chained-transfer obstruction: proof note

Status: mathematical note, step 5 of `research/chained_prime_transfer_20261111/NEXT_STEPS_20261112.md`. It gives an argument for or a citation of each step of the obstruction, marks assumptions, and ties every step to the code or measurement in this repository that checked it. It settles nothing about cross-characteristic maps (section 5).

Every number that comes from an experiment carries `[pcc:unverified]`: no pcc claim has been signed for any of them. Theorems proved or cited here are not experimental claims.

Knowledge-base step. `scripts/research-kb` was run and failed with "no executable found; set RESEARCH_KB_BIN or put research-kb on PATH". No `source_id` was obtained, and no novelty or "not tried" claim is made here. The literature is cited from the author's knowledge; theorem and section numbers marked (n) are from memory and should be checked against the source before the note is quoted.

Notation. p is a prime, $k = \mathbb{F}_p$, $\bar{k}$ an algebraic closure. E/k is an elliptic curve with origin 0; $G \in E(k)$ has prime order r; $T = [x]G$. C/k is a smooth projective geometrically irreducible curve of genus g; $c: C \to E$ is a finite morphism over k of degree d. $J_C = \mathrm{Pic}^0(C)$. c^* and c_* are pull-back and push-forward on divisors and, by functoriality, on Pic^0.

0. Summary of statuses

#	Statement	Status
T1	$c_* c^* = [d]$ on E	proved (divisor level, standard)
T2	$J_C \sim E \times P$, $\dim P = g-1$, $\#J_C(\mathbb{F}_p) = \#E(\mathbb{F}_p)\cdot\#P(\mathbb{F}_p)$	cited (Poincaré reducibility, Tate) plus short proof
T3	$\#P(\mathbb{F}_p) \in [(\sqrt{p}-1)^{2(g-1)}, (\sqrt{p}+1)^{2(g-1)}]$	proved from Weil's bound (cited)
T4	c^* is injective on the prime-to-d part, so the DLP of order $r \nmid d$ transfers	proved
T5	Fibre rigidity: a reduced divisor of weight $\le g$ in the	proved

#	Statement	Status
	class of `c^*Q` is the fibre `c^*Q`, for all `Q` outside an explicit finite set (hyperelliptic `C`, `d ≤ g`, hypothesis H0)	
T6	Same conclusion for non-hyperelliptic `C` when `g > d² − d + 1` and `c` is indecomposable	proved from Castelnuovo–Severi (cited)
T7	Collision equivalence: every dependency among fibre relations is a collision `Q_i = ±Q_j` on `E`	proved from T5 and H2
T8	`φ2_* φ1^* = 0` for the two quotients of a bielliptic curve	proved
T9	Isogenies, pairings and Weil restriction keep the characteristic; isogenies keep `#E`	cited
O1	No efficient cross-characteristic DLP-preserving map exists	open, not claimed
O2	Lifting to characteristic 0 beats C_0/p	open; the only bound is conditional
O3	An IC that draws relations from all of `J_C(F_p)` rather than from the transferred subgroup	not covered by T5–T7; known cost is worse, but that is a complexity comparison, not a theorem here

1. The cover decomposes the Jacobian; pull-back preserves the order-`r` DLP

T1 (`c_* c^* = [d]`). For every divisor `D` on `E`, `c_* c^* D = d·D`.

Proof. By linearity take `D = P` a closed point. `c^*P = Σ_{Q ↦ P} e_Q Q` where `e_Q` is the ramification index of `Q` over `P`, and `c_* Q = f_Q P` with `f_Q` the residue degree. So `c_* c^* P = (Σ_Q e_Q f_Q) P = d·P` by the fundamental identity `Σ e_Q f_Q = d` for a finite morphism of smooth curves (Hartshorne, *Algebraic Geometry*, II.6, proof of Prop. 6.9; Stichtenoth, *Algebraic Function Fields and Codes*, Thm 3.1.11 (n); Silverman, *AEC*, Prop. II.3.6 (n)). Both maps respect linear equivalence, so the identity holds on $\mathrm{Pic}^0(E) = E$. ∎

Code check: `Bielliptic::push` applied to `Bielliptic::pull(Q)` equals `[2]Q` and the cross composite is zero, on arbitrary points at `p ∈ {101, 211, 1009}` (`bielliptic.rs`, test `push_pull_identities_on_arbitrary_divisors`); `pushforward.rs`, test `pullback_pushes_to_double_and_cross_composite_vanishes`; `pushforward_deg3.rs`, test `push_pull_is_multiplication_by_three`. The degree-3 binary also checked the pull-back homomorphism for every `k ∈ 1..r` in each of 24 instances (854 checks) [pcc:unverified].

T2 (decomposition). Write `ε = c^* c_* ∈ End(J_C)`. By T1, $ε^2 = c^*(c_* c^*)c_* = d·ε$. Put $P = (\ker c_*)^0$, an abelian subvariety over `k` (`c_*` is defined over `k`). Since `c_* c^* = [d]` is surjective, `c_*` is surjective onto `E`, so `dim ker c_* = g − 1` and `dim P = g − 1`. The addition map `μ: E × P → J_C, (e, x) ↦ c^* e + x`, is defined over `k`. Its kernel is `{(e, −c^* e) : c^* e ∈ P}`. For such `e`, `c_* c^* e = d e = 0`, so the kernel is finite, contained in `E[d]` in its first coordinate. A finite-kernel homomorphism between abelian varieties of equal dimension is surjective, so `μ` is an isogeny over `k`:

> `J_C ~ E × P` over `F_p`, with `ker μ ⊂ E[d] × P[d]`.

This is Poincaré's complete reducibility theorem applied to the abelian subvariety `c^*(E)`, made explicit with the norm endomorphism `ε/d` (Milne, *Abelian Varieties*, section on Poincaré's complete reducibility; Birkenhake–Lange, *Complex Abelian Varieties*, Ch. 5 §3 and Ch. 12 on norm endomorphisms and Pryms (n)). Isogenous abelian varieties over `F_p` have equal Frobenius characteristic polynomials, hence equal point counts (Tate, *Endomorphisms of abelian varieties over finite fields*, Invent. Math. 2 (1966), Thm 1). So

> `#J_C(F_p) = #E(F_p) · #P(F_p).`

Code check: for genus 2 bielliptic curves the library computes `jacobian_order() = e1.order() * e2.order()` (`bielliptic.rs`, same test, last assertion), consistent with `P ~ E2` [pcc:unverified]. For genus 2, `J_C ~ E1 × E2` for a bielliptic curve is classical (Kani–Rosen, *Idempotent relations and factors of Jacobians*, Math. Ann. 284 (1989)).

T3 (size of P). Let `α_1, …, α_{2(g−1)}` be the Frobenius eigenvalues of `P`. Weil's Riemann hypothesis for abelian varieties gives `|α_i| = √p` (Weil 1948; Milne, *Abelian Varieties*, on the Riemann hypothesis; for `E` it is Hasse's bound). Then `#P(F_p) = Π(1 − α_i)` and `(√p − 1)^{2(g−1)} ≤ #P(F_p) ≤ (√p + 1)^{2(g−1)}`, that is

> `#P(F_p) = p^{g−1} (1 + O_g(p^{−1/2})).` ∎

Consequence: `#J_C(F_p) ≈ p^g`, of which only the factor `#E(F_p) ≈ p` carries the instance (see T4). A DLP of order `r ≤ #E(F_p)` has generic cost `Θ(√r)`; embedding it in a group of order `≈ p^g` does not shrink it.

T4 (pull-back preserves the order-r DLP). Suppose `r ∤ d`. Then `c^*` is injective on `E[r]`: if `c^* Q = 0` with `Q ∈ E[r]`, then `dQ = c_* c^* Q = 0`, and `gcd(r, d) = 1` gives `Q = 0`. So `D := c^*(G − O)` has order `r` in `J_C(F_p)`, and `c^*([x]G − O) = [x]D`: the map `⟨G⟩ → ⟨D⟩` is a group isomorphism computable by one fibre computation (a degree-`d` polynomial factoring or root-finding over `F_p` plus Cantor reduction). In general `ker c^* ⊂ E[d]`, and `ker c^* = 0` iff `c` does not factor through a nontrivial isogeny onto `E` (the classical criterion for complex curves, Birkenhake–Lange Ch. 11–12 (n); it is not needed below because `r ∤ d`). ∎

Code check: `Hop::pullback` runs `check_dlp` on construction (order of the image equals `r`, `h([x]P) = [x]h(P)` for sampled `x`) and rejects the hop otherwise (`chain.rs`). The hop's comment records the same argument for odd `r`.

2. The transferred subgroup lies in `c^*(E)`, and its relations are fibres

The transferred subgroup is `S = c^*(E(F_p)) ⊂ J_C(F_p)`. An index-calculus (IC) relation for the instance `(D, T = [x]D)` is an effective rational divisor `Σ m_P P` supported on a factor base of places such that `[Σ m_P P − w ∞_C] = [α]D + [β]T ∈ S`, with `w = Σ m_P` and `∞_C` a rational reference point. The measurements ask whether such relations carry anything beyond the point `Q = [α]G + [β]T_E ∈ E(F_p)`.

Hypotheses.

- **H0** (reference). `c^*(O) ~ d·∞_C`. In the degree-3 experiment `c^*(O) = 3∞_C` as divisors (`c` is totally ramified over `0`). In the bielliptic experiment `c^*(O) = ∞⁺ + ∞⁻`, two hyperelliptic conjugates, which is linearly equivalent to `2·(Weierstrass point)`, the reference of the quintic model (`bielliptic.rs`, module comment).
- **H1** (hyperelliptic). `C: y² = f(x)` with `∞_C` a rational Weierstrass point at infinity of the imaginary model and `d ≤ g`. This holds in both experiments (genus 2 with `d = 2`, genus 4 with `d = 3`).
- **H2** (sign). `c ∘ ι_h = [−1] ∘ c`, where `ι_h` is the hyperelliptic involution. This holds for `φ1` and for `c(x, y) = (h(x), y)`.

A divisor $\Sigma_{i\le w}\ P_i$ of weight $w \le g$ is *reduced* if it avoids ∞_C, contains no pair $P, \iota_h P$ (counting a Weierstrass point at most once).

Mumford–Cantor uniqueness (Mumford, *Tata Lectures on Theta II*, IIIa §2; Cantor, Math. Comp. 48 (1987)). For hyperelliptic C with rational Weierstrass point ∞_C, every class in $J_C(k)$ is $[\Sigma\ P_i - w\ \infty_C]$ for a unique reduced rational divisor $\Sigma\ P_i$ of weight $w \le g$.

T5 (fibre rigidity). Assume H0 and H1. Let $Z = \{P \in C(\bar{k}) : c(\iota_h P) = c(P)\}$, let Ram be the ramification points of c, and $N = c(Z) \cup \{c(\infty_C)\} \cup c(\mathrm{Ram} \cap \mathrm{Weier})$ where Weier is the set of Weierstrass points. Then N is finite, and for every $Q \in E(k) \setminus N$ the reduced representative of the class $c^*[Q - O]$ is the fibre c^*Q itself, of weight exactly d. Consequently, every IC relation of weight $\le g$ whose class is $c^*[Q - O]$ is the full fibre c^*Q, with each place P appearing with its ramification multiplicity $e_P \le d$.

Proof. Finiteness. Ram and Weier are finite. Suppose Z were infinite. Then $c \circ \iota_h$ and c agree on infinitely many points, hence everywhere, so c is constant on ι_h-orbits and factors through $C/\iota_h = P^1$, which is impossible, since every morphism $P^1 \to E$ is constant. So Z is finite (it is the zero set of the nonconstant morphism $P \mapsto c(P) - c(\iota_h P)$ to E, which is nonconstant: were it a constant e_0, applying ι_h gives $2e_0 = 0$, and $\pi : E \to E/\langle e_0\rangle$ would satisfy $\pi\ c\ \iota_h = \pi\ c$, so $\pi\ c$ would factor through P^1 and be constant, which is impossible).

Reducedness. Let $Q \notin N$. The effective divisor c^*Q has degree $d \le g$. By H0, $[c^*Q - d\infty_C] = [c^*Q - c^*O] = c^*[Q - O]$. The divisor c^*Q avoids ∞_C (else $Q = c(\infty_C) \in N$), contains no pair $P, \iota_h P$ (else $c(P) = c(\iota_h P) = Q$, so $Q \in c(Z)$), and has no Weierstrass point with multiplicity ≥ 2 (else c ramifies there). So it is reduced, and by Mumford–Cantor uniqueness it is *the* reduced representative of the class. A relation of weight $\le g$ with that class is a reduced divisor (the decomposition of the reduced representative over the factor base), hence equals c^*Q. ∎

Where it can fail. (a) H0 fails: the class of the relation is then $c^*[Q - O]$ shifted by $\kappa = [c^*O - d\infty_C] \ne 0$, and the divisor $c^*Q + (d\infty_C - c^*O)$ has no reason to be a fibre. (b) $Q \in N$: finitely many points, among them $c(\infty_C)$; for such Q the reduced representative is shorter than d, and the relation is not a full fibre. (c) Weight $> g$: a relation with $m > g$ places has a linear system of dimension $\ge m - g$, with many non-fibre members; T5 does not constrain it, and the standard IC does not use such relations. (d) $d > g$ or non-hyperelliptic C: use T6. (e) The relation is drawn from outside S: see O3.

T6 (non-hyperelliptic version via Castelnuovo–Severi). Let C be any curve, $c : C \to E$ of degree d over $\bar{k}$, and suppose

- $g > d^2 - d + 1$, and
- c is indecomposable: it does not factor as $C \to C' \to E$ with $C' \ne C$ and $C' \ne E$ (in the sense of both degrees > 1).

Then $\ell(c^*Q) = 1$ for every $Q \in E(\bar{k})$; so any effective divisor D with $D \sim c^*Q$ equals c^*Q.

Proof. Suppose $\ell(c^*Q) \ge 2$. Two independent sections give a nonconstant $f : C \to P^1$ of degree $e \le d$. Let $K_c = \bar{k}(E)$ and $K_f = \bar{k}(f)$, subfields of $\bar{k}(C)$ of indices d and e. If their compositum is $\bar{k}(C)$, the Castelnuovo–Severi inequality (Stichtenoth, Thm 3.11.3 (n)) gives $g \le d\cdot 1 + e\cdot 0 + (d-1)(e-1) \le d + (d-1)^2 = d^2 - d + 1$, contradicting the hypothesis. Otherwise the compositum is $\bar{k}(C'')$ for a curve C'' with $C \to C''$ of degree > 1, and c factors through C''. By indecomposability $C'' \cong E$, so $f = u \circ c$ with $u : E \to P^1$ nonconstant of degree ≥ 2, and $e = d\cdot\deg u \ge 2d > d$, a contradiction. ∎

T6 does not apply to either experiment ($g = 2, d = 2$ needs $g > 3$; $g = 4, d = 3$ needs $g > 7$). It is the cleanest criterion for the non-hyperelliptic setting and shows the failure mode is a pencil, not an arithmetic accident.

Two sharper facts for the genus-2 bielliptic case, $d = 2$, $g = 2$: a degree-2 effective divisor on a genus-2 curve moves iff it is canonical, so $\ell(c^*Q) = 2$ iff $c^*Q \sim K_C$. Here $c^*O = \infty^+ + \infty^- \sim K_C$, so $Q = O$ is a point with a moving fibre, and the identity class is not a relation anyway. If $c^*Q \sim K_C$ and $c^*Q' \sim K_C$ then $Q - Q' \in \ker c^* \subset E[2]$; the

kernel meets `E × 0` trivially in Kani's description of `J_C` as the quotient of `E1 × E2` by the graph of an isomorphism `E1[2] ≅ E2[2]`, so `Q = 0` is the only moving fibre. (The kernel claim is cited from Kani–Rosen and the description of split genus-2 Jacobians, Howe–Leprévost–Poonen, Forum Math. 12 (2000) (n); it is not used by T5.)

T7 (collision equivalence). Assume H0, H1, H2, and `r > d` prime. Let the IC collect rows `i = 1, …, n` with `Q_i = [α_i]G + [β_i]T_E ∉ N`, each row being the fibre `c^*Q_i` written on the factor-base columns, where a place `P` and `ι_h P` are identified up to sign (as the code identifies places with `y ≤ (p−1)/2`). Then:

1. The columns supporting the row of `Q_i` are exactly the places of `c^{-1}(Q_i)` and their `ι_h`-images, which by H2 are the places of `c^{-1}(−Q_i)`. Rows with `Q_i ≠ ±Q_j` have disjoint column supports.
2. Every nonzero entry of a row is a ramification multiplicity in `1..d`, a unit mod `r`. So the rows with distinct classes `±Q` are linearly independent over `Z/r`, and rows with `Q_j = ±Q_i` are `±1` multiples of one another.
3. Every dependency `Σ λ_i row_i = 0` over `Z/r` is therefore a sum of pairwise dependencies between rows with `Q_j = ε Q_i`, `ε = ±1`. Such a pair gives `(α_i − εα_j) + (β_i − εβ_j) x ≡ 0 (mod r)`, which is exactly the statement `Q_i = ε Q_j` on `E`.
4. Hence the first dependency, and any logarithm the IC extracts, requires a repeated `±Q` among the accepted trials. The accepted trials are a subsequence of the unfiltered stream `Q_1, Q_2, …` on `E`, so on the same random stream the first repeat of an unfiltered birthday search on `E` occurs no later. ∎

Reading. Under these hypotheses, index calculus restricted to the transferred subgroup, over any factor base of places of `C`, is a birthday collision search on `E` with an acceptance filter ("the fibre `c^{-1}(Q)` factors over the allowed places") that can only discard trials. The same holds if the factor base contains places of degree `> 1` (large primes): a fibre that factors into places of several degrees is still the single fibre over a single `Q ∈ E(F_p)`.

Measurements that tested T5 and T7 [pcc:unverified]:

- `research/cover_pushforward_20261112/PUSHFORWARD.md` (degree 2, genus 2, 7 primes 211 to 12841, 3 seeds, plus a 15-seed run): 37,866 of 37,866 collected relations are fibre type; 189,330 of 189,330 in the 15-seed run; 0 invalid pushed relations on `E1` or `E2`; columns used by the genus-2 arm are exactly twice those of the native `E1` arm (each place `P` and `ι P` over the same `X`); the control arm A6 (birthday search with no filter) finds its first collision at or before the cover arm in every cell. Code: `pushforward.rs`, `collect_jc`, `push_check`, `solve_rows`.
- `research/cover_pushforward_deg3_20261112/PUSHFORWARD_DEG3.md` (degree 3, genus 4, 8 primes 53 to 251, 24 cells, 360 seed-cells): 28,815 of 28,815 relations are a single full fibre (0 unions, 0 other), including 1,317 rows with a repeated root; in 159,508 of 159,508 trials the Cantor-reduced class equals the Mumford pair of `Q` computed on `E` alone; the J(C) arm and the native `E` arm have identical counted operations and recover the same log in 360 of 360 seed-cells. Code: `pushforward_deg3.rs`.

These counts are consistent with T5 and T7 but are not their justification; T5 and T7 have their own arguments above, and the counts check that the code implements the setting of the proof (hypotheses H0 to H2, `d ≤ g`, weight `≤ g`).

Consequence stated once. For the instance transferred along `c`, an IC that takes its relations in `S` and in reduced weight `≤ g` cannot do better than the matched collision search on `E`, up to the discarded trials. This does not exclude other relation sources (O3) nor any method that is not an IC.

3. The complement receives nothing: φ2_* φ1^* = 0

Setting: `C: y² = F(x²)`, `σ(x, y) = (−x, y)`, `φ1(x, y) = (x², y)` to `E1: Y² = F(X)`, and `φ2(x, y) = (1/x², y/x³)` to `E2: Y² = X³F(1/X)`. Then `φ1 ∘ σ = φ1` and `φ2 ∘ σ = [−1] ∘ φ2`, the latter because `φ2(−x, y) = (1/x², −y/x³)` is the negation on `E2`.

T8. φ2_* φ1^* = 0 as a homomorphism E1 → E2.

Proof. Put ψ = φ2_* φ1^*. From φ1 σ = φ1 we get σ_* φ1^* = φ1^* (pull-back by φ1 is σ-invariant, and σ is an involution, so σ^* = σ_* on J_C). From φ2 σ = [−1] φ2 we get φ2_* σ_* = [−1] φ2_*. Therefore ψ = φ2_* φ1^* = φ2_* σ_* φ1^* = −φ2_* φ1^* = −ψ, so 2ψ = 0 and ψ(E1) ⊂ E2[2]. The image of a connected variety under a morphism is connected; a finite connected set is a point; so ψ = 0. ∎

Reading. S ⊂ φ1^*(E1) is invisible to E2: any relation pushed to E2 is Σ = 0 whatever (α, β) are, so nothing about the target is encoded there. For a general cover c, c_* kills P by definition, and c_* c^* = [d] shows c_* does not kill c^*(E), so the same conclusion holds for P as a whole: the target-carrying part is the E-isotypic part.

Code and measurement: bielliptic.rs asserts b.push(other, &d) == None (identity) for the pull-back of generic points of both quotients, on p ∈ {101, 211, 1009}. PUSHFORWARD.md: φ2_*(G) and φ2_*(T) are the identity in 37,866 of 37,866 rows and 0 pushed E2 rows carry the target [pcc:unverified].

4. Isogenies, pairings and Weil restriction keep the characteristic

- **Isogenies.** An F_p-isogeny φ: E → E' has #E(F_p) = #E'(F_p) and conversely two curves over F_p with the same number of points are F_p-isogenous (Tate, Invent. Math. 2 (1966), Thm 1; Silverman, *AEC*, Ch. V and Exercise 5.4 (n)). The order r and the embedding degree k = min{k : r | p^k − 1} depend only on (p, r), so an isogeny keeps both. An isogeny transfers the DLP exactly when ker φ ∩ ⟨G⟩ = 0, in particular when r |/ deg φ.
- **Precisely what the Tate statement gives.** An algebraic group homomorphism E → E' over F_p that is nonzero is an isogeny, so it can exist only if #E = #E'. That direction is established by the cited theorem. The converse gives an isogeny, not an isogeny of degree prime to r; I do not claim that one exists in every isogeny class. In particular, “DLP-preserving maps exist iff #E = #E'” (as the wording of ATLAS.md and the earlier notes reads) is true only for algebraic homomorphisms, and its “if” direction is not shown by Tate’s theorem alone. Set-theoretic isomorphisms of cyclic groups of order r always exist and are not efficiently computable.
- **Into higher genus.** If ψ: E → A is an F_p-homomorphism with finite kernel, ψ(E) is an elliptic curve isogenous to E, so the image of ⟨G⟩ sits in a group of order #E(F_p) (Tate for abelian varieties). This is the reason the transferred subgroup lives in the E-factor of T2.
- **Pairings.** The Menezes–Okamoto–Vanstone reduction (IEEE Trans. IT 39 (1993)) and the Frey–Rück reduction (Math. Comp. 62 (1994)) embed ⟨G⟩ into μ_r ⊂ F_{p^k}^* via the Weil or Tate pairing. The target is a multiplicative group of a field of characteristic p and of size ≈ p^k. For a random curve k is of the order of r (Balasubramanian–Koblitz, J. Cryptology 11 (1998)), so the reduction is useful only for small k.
- **Weil restriction and descent.** Weil restriction Res_{F_{p^n}/F_p} (Frey 1998; Gaudry–Hess–Smart, J. Cryptology 15 (2002)) changes the base field but not its characteristic. A curve over the prime field F_p has no proper subfield to descend to, so the construction is vacuous there. Descent for extension fields is a separate question and its known limits are surveyed by Diem, Compositio Math. 147 (2011) (n).
- **Smart’s anomalous attack** (J. Cryptology 12 (1999)) works by lifting to Q_p and is available only when #E(F_p) = p; it does not apply to a curve of other trace. It shows that the characteristic-0 route is real but special.

Code and measurement: chain.rs (Hop::isogeny) checks order and the homomorphism property for each Vélu isogeny hop and rejects it otherwise; the reachable-set census is in atlas.rs and research/chained_prime_transfer_20261111/ATLAS.md [pcc:unverified].

5. What is not established

- **O1.** That no efficient cross-characteristic map from ⟨G⟩ to a group in which the DLP is easier exists is open. T1 to T9 only show that the maps built here (covers, isogenies, pairings, Weil restriction) stay in characteristic p and carry the instance in the E-factor. They are not a barrier theorem against an unknown map. Nothing in this repository constructs a cross-characteristic map.
- **O2.** The only known route out of characteristic p, lifting to characteristic 0 (the xedni calculus, Silverman 1999), has a success bound of about C_0/p per trial in Jacobson–Koblitz–Silverman–Stein–Teske, *Analysis of the Xedni Calculus Attack*, Designs, Codes and Cryptography 20 (2000), 41–64, Thm 4.1, which begins "under certain plausible assumptions" and depends on a height-versus-discriminant comparison and Lang's height conjecture. It is conditional. The repository's Lean audit (`research/jksst_xedni_audit_20261101/`, `PROOF_MAP.md`) proves the conditional theorem from its stated hypotheses and refutes several overbroad variants; it does not discharge Lang's conjecture, and it does not show the algorithm fails.
- **O3.** T5 to T7 apply to relations in the transferred subgroup S. An IC in the style of Gaudry (Eurocrypt 2000) draws relations from random elements of the whole J_C(F_p) and computes logarithms of factor-base elements in the full group, of order ≈ p^g (T3). Its known cost is Õ(p^{2−2/g}) (Gaudry–Thomé–Thériault–Diem, Math. Comp. 76 (2007)), against √r ≤ √p for rho on the target subgroup: worse for every g ≥ 2. That is a comparison of best-known algorithms, not a lower bound, and the `tower_large` and `hyper64.rs` notes measure it only at toy size.
- **Measured results are toy-size.** Every experiment cited has p at most 12,841 (degree 2) or 251 (degree 3), a handful of covers per prime, and no signed pcc claim: they are `[pcc:unverified]`, and none is a row in the validation gate. They check that the code implements the hypotheses of the proofs; they do not extend the proofs to larger p.
- T5 to T7 do not cover non-hyperelliptic covers with $g \le d^2 - d + 1$ (T6 is silent), relations of weight > g, references with κ ≠ 0, or the finite exceptional set N.

6. Claims in earlier notes that this note narrows

1. `NEXT_STEPS_20261112.md`, step 5: "effective divisors of degree deg c in c^*(E) are fibres". Established only under H0 plus either T5 (hyperelliptic C, d ≤ g, weight ≤ g, Q ∉ N) or T6 ($g > d^2 - d + 1$, indecomposable c). For non-hyperelliptic curves of small genus and for Q = 0 in the bielliptic case it can fail.
2. `ATLAS.md`: "a DLP-preserving homomorphism E → E' exists iff E and E' are isogenous, and over F_p iff #E = #E'". The "only if" is shown above. The "if" gives an isogeny whose degree may be divisible by r (section 4).
3. `PUSHFORWARD.md`: "the reduced class of [α]G + [β]T is a fibre" holds for Q ∉ N; the finite set N (fibres through ∞_C, through a conjugate pair, or through a ramified Weierstrass point) is excluded, and the Q = 0 class is the identity. This matches `Bielliptic::fiber_generic` in the code, which excludes the ramification locus.
4. "IC on the transferred subgroup is a collision search on E" is a statement about relations taken in S (O3). It is not a statement about IC on all of J_C.
5. c^* is injective only on the prime-to-d part in general (T4); in the bielliptic experiment it is injective for the odd r used.

7. Lean candidates

Search results. `/home/user/crypto` has Lean projects, e.g. `research/jksst_xedni_audit_20261101/lean/lakefile.toml` (toolchain `leanprover/lean4:v4.34.1`), `research/prime_square_kernel_20261106/lean/lakefile.toml`, and the `research/koblitz_modular_ic_20261018/rounds/*/lean/` projects. `lake` and `lean` are not on `PATH` in this container, and no built Mathlib is present (only a partial source cache under `research/pencil_lifecycle_20261030/.../cache-sources/Mathlib`, without

RamificationInertia). /home/user holds only crypto and cryptanalysis; crypto-lean is not available locally. So **the statements below are text only and nothing here was type-checked.**

The divisor-level core of T1 can be stated without algebraic geometry beyond the fundamental identity. Candidate formulation (abstract, to be instantiated by a Dedekind-domain extension, where Mathlib has the fundamental identity Ideal.sum_ramification_inertia (n)):

```
-- c_* c^* = [deg c] at the level of divisors (free abelian groups on
places).
-- Hypothesis `fund`: for every place P of E, Σ_{Q | P} e(Q|P) f(Q|P) = d.
theorem pushforward_pullback_eq_smul
    {Pl_C Pl_E : Type*} [DecidableEq Pl_C] [DecidableEq Pl_E]
    (over : Pl_C → Pl_E) (e f : Pl_C → ℕ) (d : ℕ)
    (fund : ∀ P : Pl_E, ∑ Q ∈ (Finset.univ.filter fun Q => over Q = P), e Q
* f Q = d) :
    ∀ D : Pl_E →₀ ℤ, pushDiv over f (pullDiv over e D) = d • D := by
  sorry
```

where pullDiv over e D = Σ_P D P • Σ_{Q | P} e Q • [Q] and pushDiv over f Q' = Σ f Q • [over Q]. The statement is a finite sum interchange once fund is a hypothesis; the geometric content (that fund holds for a finite morphism of smooth curves) is the cited fundamental identity. A second formulation, on class groups, would add that both maps respect principal divisors; Mathlib lacks function fields of curves with Pic.

The bielliptic cross-map vanishing (T8) in abstract form: for abelian groups A (standing for J_C), B1, B2, homomorphisms a: B1 → A, b: A → B2, and an automorphism s: A → A with s ∘ a = a and b ∘ s = −b, then b ∘ a = −(b ∘ a), that is 2 • (b ∘ a) = 0:

```
theorem cross_map_two_torsion
    {A B1 B2 : Type*} [AddCommGroup A] [AddCommGroup B1] [AddCommGroup B2]
    (a : B1 →+ A) (b : A →+ B2) (s : A →+ A)
    (h1 : ∀ x, s (a x) = a x) (h2 : ∀ y, b (s y) = - b y) :
    ∀ x, b (a x) + b (a x) = 0 := by
  intro x; have := h2 (a x); rw [h1] at this; ...  -- b (a x) = - b (a x)
  sorry
```

The upgrade from 2ψ = 0 to ψ = 0 uses connectedness of E1 (a continuous image of a connected variety is connected), which has no existing Lean formulation for abelian varieties; the algebraic statement above is the “Lean-ready” part. The T4 step (c^* is injective on E[r] for gcd(r, d) = 1) is a one-line group-theory lemma from c_* c^* = [d]: ∀ x, r • x = 0 → c_pull x = 0 → d • x = 0 → x = 0 for coprime r, d.

8. Files and verification

- This note: research/chained_transfer_proof_20261112/PROOF.md; rendered copy PROOF.pdf in the same directory.
- Code referred to: src/cryptanalysis/chained_transfer/bielliptic.rs, chain.rs, pushforward.rs, pushforward_deg3.rs, atlas.rs.
- Not run: no experiment was run for this note; no test was run. The code checks cited are the repository’s existing tests and the recorded measurement notes.